

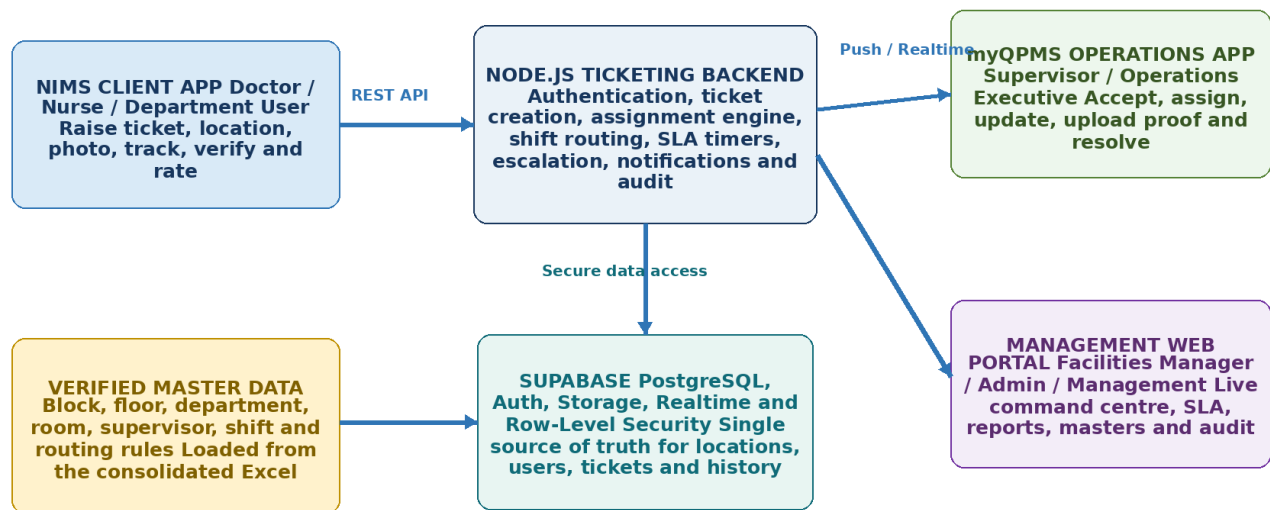
# NIMS HYDERABAD

## Client Ticketing Data Architecture

How the available block, floor, department, room, manpower and supervisor data will be used in the application

### NIMS Ticketing - System Architecture

One shared ticketing backend and database for the client app, myQPMS operations app and management portal.



### DOCUMENT STATUS

Architecture and data-use design for the NIMS Phase 1 pilot. The consolidated Excel contains Draft records; production must expose only Verified and active locations and routing rules.

Prepared for Quality Property Management Services Pvt Ltd

# 1. Executive Summary

The available data is sufficient to configure a strong NIMS pilot location master and draft automatic supervisor routing. It is not yet a complete indoor map. The application will combine structured master data with a mandatory exact landmark and photo so the supervisor can identify the physical problem even when a public service directory does not contain every corridor, washroom, nursing station or equipment point.

## Data consolidated

| Source                               | Available data                                                                                                              | How it will be used                                                   | Current confidence                                                      |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------|
| ROLES.pdf (2 pages)                  | 15 supervisors/operations roles, shifts, phone numbers, assigned areas and responsibilities                                 | Draft block/shift routing, notifications and escalation hierarchy     | Medium - overlapping responsibilities need confirmation                 |
| Document 2.pdf (11 pages)            | 238 department deployment rows, block-wise posts, 1,197 unskilled and 350 semi-skilled manpower, add/minus adjustment sheet | Block/department dropdowns, staffing analytics and coverage reference | High for listed staffing totals; floor/room detail is incomplete        |
| User-pasted online service directory | 85 morning and 19 evening service rows with floor, room/cabin, days and timings                                             | Draft Block > Floor > Department > Room indoor location master        | Medium - source date/URL not included and OCR-like text needed cleaning |

## Core decision

### PILOT LOCATION RULE

A ticket must never contain only the complaint description. It must contain Block + Floor + Department/Area + Room/Cabin where known + mandatory Exact Landmark + photo. The structured path gets the supervisor to the correct area; the landmark and photo identify the final spot.

## Production gate

- Only Verified location and routing records should be visible to live NIMS users.
- The app may use Draft records in a controlled test environment.
- Block aliases and overlaps must be resolved: SPL/Speciality, MIL/Millennium, OPD/Admin/Old Building, and Emergency & Physiotherapy versus Trauma/EMD.
- Supervisor primary, backup, weekly-off and leave-cover rules must be formally confirmed.

## 2. Data Foundation and Governance

### From Raw Documents to Live App Master Data



**Control principle:** a record can be used for testing while Draft, but it must not be visible to live NIMS users until it is Verified and activated.

### Master-data layers

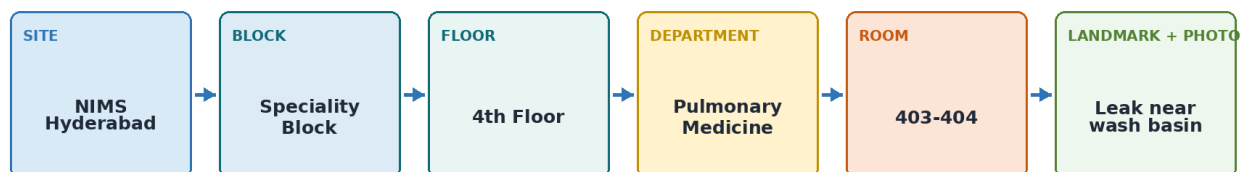
| Layer                    | Examples                                                                                                                   | Owner                                | Live-app rule                                                                  |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------|
| Site master              | NIMS Hyderabad                                                                                                             | QPMS Admin                           | One active pilot site                                                          |
| Block master             | Admin, Core, Millennium, Oncology, Speciality, Trauma, Extra Mural; additional OPD/Emergency labels pending reconciliation | QPMS + NIMS Administration           | Canonical names only; aliases retained for search                              |
| Floor master             | Ground, First, Second, Third, 4th, 5th, 6th and 7th where explicitly mentioned                                             | NIMS Engineering / Block Supervisor  | Create verified floors; a mentioned floor does not prove total building height |
| Department/area master   | Ward, OPD, ICU, OT, lab, washroom, billing, waiting hall and support areas                                                 | Block Supervisor                     | Linked to block and optional floor                                             |
| Exact location master    | Room/cabin, nursing station, washroom ID, corridor, lift/staircase and QR code                                             | Block Supervisor / QPMS Project Team | Highest-precision selectable location                                          |
| Staff and routing master | Supervisor, shift, assigned blocks, primary/backup and escalation                                                          | QPMS Facilities Manager              | Only verified effective assignments used by backend                            |

### Verification status model

| Status       | Meaning                                                                | App behaviour                                     |
|--------------|------------------------------------------------------------------------|---------------------------------------------------|
| Draft        | Imported or cleaned from available documents; not yet confirmed onsite | Visible only to admin/test users                  |
| Needs Review | Conflict, alias, missing floor/room or overlapping routing exists      | Highlighted for responsible reviewer              |
| Verified     | Confirmed by authorised NIMS/QPMS reviewer                             | Eligible for activation                           |
| Inactive     | Old, relocated or temporarily unavailable location/staff assignment    | Hidden from new ticket creation; history retained |

### 3. Location Architecture - How the Supervisor Finds the Problem

#### How the Supervisor Gets the Exact Problem Location



The PDF/Excel data can prefill Block, Floor, Department and Room. The app must still require Exact Landmark and a photo because public directories do not map every corridor, washroom, bed zone or equipment point.

#### Client ticket form behaviour

| Step | Field                          | System behaviour                                                                                  | Required?                  |
|------|--------------------------------|---------------------------------------------------------------------------------------------------|----------------------------|
| 1    | Site                           | Defaults to NIMS Hyderabad for the pilot                                                          | Yes                        |
| 2    | Block                          | Loads from verified block master                                                                  | Yes                        |
| 3    | Floor                          | Filtered by selected block; allow Not Known only during controlled pilot                          | Yes / controlled fallback  |
| 4    | Department / Ward / Area       | Filtered by block and floor                                                                       | Yes                        |
| 5    | Room / Cabin / Sub-area        | Auto-suggested where known                                                                        | Optional when unavailable  |
| 6    | Exact Landmark                 | Free text such as “washroom opposite Bed 12” or “corridor near Lift 2”                            | Yes                        |
| 7    | Issue category and description | Housekeeping, security, electrical, plumbing, civil, biomedical, HVAC, lift and other FM services | Yes                        |
| 8    | Photo                          | Shows the issue and surrounding landmark; category may make it mandatory                          | Recommended / configurable |

#### Example ticket shown to supervisor

##### **NIMS-SPL-20260721-0048**

Speciality Block > 4th Floor > Pulmonary Medicine > Rooms 403-404

Exact landmark: Wash basin in corridor outside Room 404

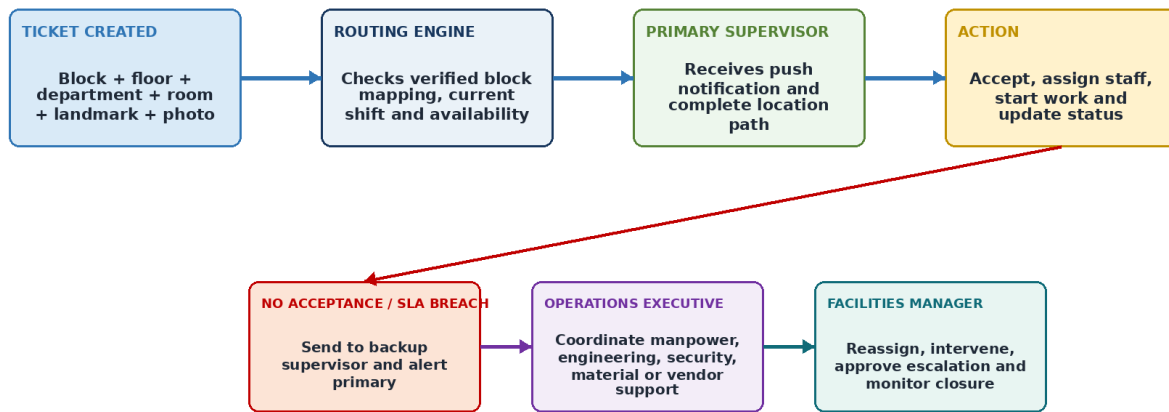
Issue: Water leakage | Priority: High | Photo attached

## Why GPS alone is not enough

GPS can confirm that the user is near the NIMS campus, but it cannot reliably distinguish floors, adjacent wards, rooms or washrooms inside a hospital. Indoor identification therefore depends on the verified location hierarchy, exact landmark, photo and later QR codes for high-volume locations.

## 4. Supervisor Routing, Notifications and Shift Handover

### Automatic Supervisor Assignment and Escalation



Important: the current roster contains overlapping coverage. The Excel routing sheet is intentionally marked Needs Review until primary, backup, weekly-off and leave-cover rules are confirmed.

### Assignment key

#### ROUTING FORMULA

Verified Ticket Block/Area + Current Shift + Active Primary Assignment + Availability = Assigned Supervisor. If no valid primary exists, use the verified backup and alert the Operations Executive.

### Draft coverage inferred from ROLES.pdf

| Area                   | Day / evening coverage visible in roster                           | Night coverage                   | Status                              |
|------------------------|--------------------------------------------------------------------|----------------------------------|-------------------------------------|
| OPD / Admin / Oncology | L. V. Sai; Ch Ramu and K Srinivas provide overall/backup oversight | Venkata Krishna Reddy / Y Nikhil | Needs Review                        |
| Speciality Block       | M. Praveen (8 AM-4 PM), Sastri (8 AM-4 PM), Shiva (2 PM-8 PM)      | Venkata Krishna Reddy / Y Nikhil | Needs Review - overlap              |
| Trauma Block           | B. Anil (7 AM-3 PM), Sastri (8 AM-4 PM), Shiva (2 PM-8 PM)         | Venkata Krishna Reddy / Y Nikhil | Needs Review - overlap and handover |
| Millennium Block       | V Lenin (8 AM-4 PM), V Anji Reddy / M Srinivas (12 Noon-8 PM)      | Venkata Krishna Reddy / Y Nikhil | Needs Review - overlap              |
| NPR / Old Building     | A Ravi (7 AM-3 PM), V Anji Reddy (12 Noon-8 PM)                    | Venkata Krishna Reddy / Y Nikhil | Needs Review                        |

### Notification content

- Ticket number, priority, category and age.
- Full location path: Block > Floor > Department/Area > Room/Cabin > Exact Landmark.

- Issue photo thumbnail with tap-to-view.
- Raised-by department and contact method, subject to role-based privacy.
- Accept, Navigate/Call, Start Work, Reassign, Escalate and Add Update actions.

## Shift handover

Open tickets must not remain attached to an off-duty supervisor. Before shift end, the app will show unresolved tickets, require a handover note, assign them to the incoming verified supervisor and record acknowledgement in the ticket timeline.

## 5. Ticket Lifecycle and SLA Architecture

| Status                                                       | Who acts                     | Required evidence / rule                                           |
|--------------------------------------------------------------|------------------------------|--------------------------------------------------------------------|
| Open                                                         | System                       | Ticket ID generated; SLA starts; location and attachment validated |
| Assigned                                                     | Primary supervisor           | Push notification sent; backup identified                          |
| Accepted                                                     | Supervisor                   | Acceptance timestamp recorded                                      |
| In Progress                                                  | Supervisor / assigned staff  | Work-start timestamp, assignee and update note                     |
| On Hold / Material Pending / Vendor Pending / Client Pending | Supervisor / Executive       | Reason and next-action date are mandatory                          |
| Resolved                                                     | Supervisor                   | Action taken, completion note, after photo where required          |
| Verification Pending                                         | Client / department user     | Client confirms satisfactory completion                            |
| Closed                                                       | System / authorised user     | Rating and closure timestamp stored                                |
| Reopened                                                     | Client / authorised reviewer | Reason is mandatory; SLA resumes or restarts per policy            |

### Escalation model

| Level                           | Trigger                                                                   | Recipient                                             | Action                                                      |
|---------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------|
| Supervisor                      | New ticket                                                                | Verified on-duty primary                              | Accept and start work within configured SLA                 |
| Backup / Operations Executive   | No acceptance, SLA warning, manual support request or primary unavailable | Verified backup and Facilities & Operations Executive | Reassign, coordinate manpower/material/engineering/security |
| Facilities & Operations Manager | Continued breach, critical issue, repeated failure or dispute             | Facilities & Operations Manager                       | Intervene, approve escalation, monitor closure              |
| Management                      | Critical unresolved matter or configured final breach                     | Authorised management users                           | Review and direct final action                              |

#### SLA CONTROL

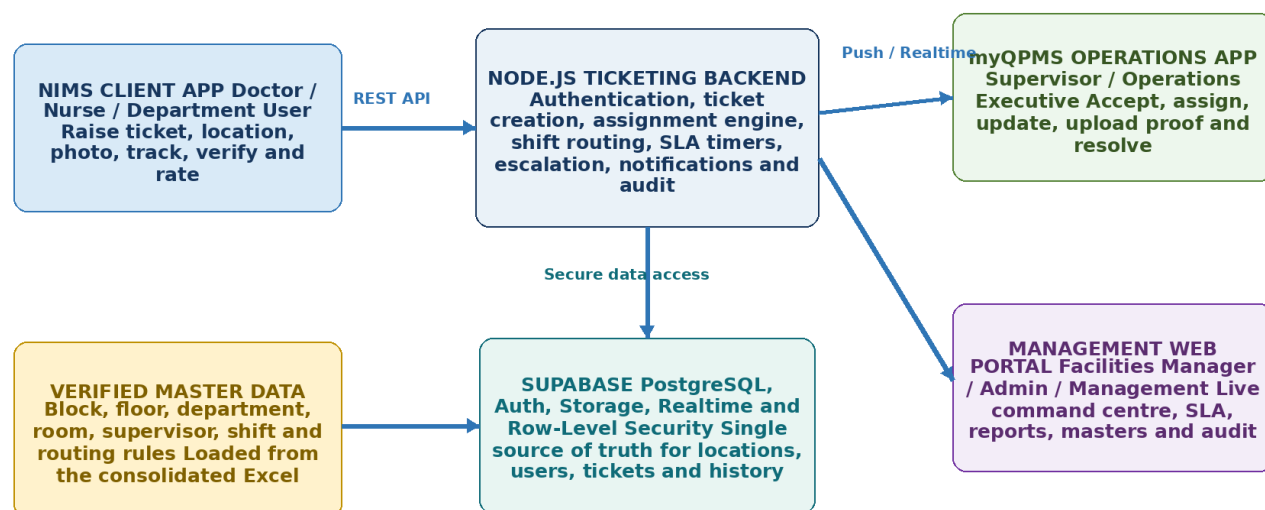
The previously discussed 20/30/40-minute escalation concept remains a draft. SLA must be configurable by priority, category, location and shift and formally approved before production.



## 6. Application and Technical Architecture

### NIMS Ticketing - System Architecture

One shared ticketing backend and database for the client app, myQPMS operations app and management portal.



### Application responsibilities

| Component                     | Primary users                                                                     | Responsibilities                                                                                                       |
|-------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| NIMS Client Flutter App       | Doctors, nurses, nursing stations, department heads and authorised hospital users | Raise complaint, select exact location, upload photo, track timeline, verify resolution, reopen and rate               |
| myQPMS Operations Flutter App | Supervisors, Facilities & Operations Executive and Manager                        | Receive assignments, accept, assign staff, update, upload evidence, resolve, hand over and escalate                    |
| React Management Portal       | Facilities Manager, QPMS management and admins                                    | Live command centre, SLA monitoring, block/category performance, user/location/routing masters, reports and audit      |
| Node.js / Express Backend     | System services                                                                   | JWT validation, ticket workflow, assignment engine, SLA scheduler, notifications, reports and protected business rules |
| Supabase                      | All apps through authorised access                                                | PostgreSQL data, Auth, file storage, realtime updates and Row-Level Security                                           |

### Security principles

- Do not hardcode supervisor phone numbers, blocks, floors or routing rules in Flutter.
- Use employee code/email-based profiles and role-based access; mobile numbers are restricted operational data.
- Client users see only their own/authorised department tickets; supervisors see assigned coverage; management sees authorised site-wide data.

- All status, assignment and master-data changes create audit records.
- Detailed architectural/as-built floor plans, if obtained, remain restricted and are not exposed to ordinary client users.

## 7. Recommended Database Model

| Table                      | Purpose                                                  | Key fields                                                                              |
|----------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------|
| hospital_sites             | Hospital/site master                                     | id, name, code, active                                                                  |
| hospital_blocks            | Canonical block master and aliases                       | id, site_id, name, code, aliases, active                                                |
| hospital_floors            | Ordered floor master                                     | id, block_id, name, floor_order, active                                                 |
| hospital_departments       | Departments, wards and operational areas                 | id, block_id, floor_id, name, category, active                                          |
| hospital_locations         | Rooms, cabins, washrooms, nursing stations and QR points | id, department_id, location_code, room_no, label, landmark, qr_code, verified_at        |
| profiles                   | Client and QPMS users                                    | id, employee_code/email, full_name, mobile, role, site_access                           |
| staff_shifts               | Shift definitions                                        | id, name, start_time, end_time, overnight                                               |
| staff_location_assignments | Effective routing rules                                  | user_id, block/department, shift_id, primary_or_backup, start_date, end_date, active    |
| tickets                    | Current ticket state and location IDs                    | ticket_number, location IDs, exact_location, category, priority, status, SLA timestamps |
| ticket_assignments         | Assignment and handover history                          | ticket_id, assigned_to, assigned_by, assignment_type, timestamps                        |
| ticket_status_history      | Immutable timeline                                       | ticket_id, from_status, to_status, actor, note, created_at                              |
| ticket_attachments         | Issue and completion evidence                            | ticket_id, storage_path, attachment_type, uploaded_by                                   |
| ticket_escalations         | SLA and manual escalation audit                          | ticket_id, level, reason, escalated_to, timestamps                                      |
| ticket_feedback            | Client satisfaction and reopen data                      | ticket_id, rating, comment, satisfied, reopen_reason                                    |

### Location ID example

#### NIMS-SPL-F4-003

Site NIMS > Speciality Block > 4th Floor > Pulmonary Medicine > Rooms 403-404. The same stable code can later be printed as a QR label.

## 8. Known Block and Floor Coverage

| Block                           | Floors explicitly mentioned in current data                                                     | Important limitation                                                                            |
|---------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| OPD Block                       | Ground Floor; First Floor                                                                       | Section heading is inferred from the pasted directory; reconcile with Admin/Old Building naming |
| Millennium Block                | Ground Floor in online directory; First, 3rd and 7th floors also referenced in deployment names | No complete floor-by-floor directory                                                            |
| Emergency & Physiotherapy Block | Ground, First and Second floors                                                                 | Confirm whether this is the same operational area as Trauma/EMD                                 |
| Speciality Block                | Ground, Third, 4th, 5th and 6th floors online; 1st and 2nd floors referenced in deployment data | This proves at least these levels, not the total floor count                                    |
| Oncology Block                  | Ground and First floors online                                                                  | No complete floor map                                                                           |
| Admin Block                     | No reliable floor directory in available documents                                              | Departments are available at block level                                                        |
| Core Block                      | 2nd and 3rd floors appear in department names                                                   | No complete floor map                                                                           |
| Trauma Block                    | First Floor appears in department names                                                         | Online directory uses a different Emergency block name                                          |
| Extra Mural                     | Campus/support areas rather than one conventional building                                      | May need separate building/zone records instead of floors                                       |

### Manpower reference

| Block            | Unskilled | Semi-skilled | Total |
|------------------|-----------|--------------|-------|
| Trauma Block     | 219       | 24           | 243   |
| Speciality Block | 202       | 33           | 235   |
| Millennium Block | 195       | 51           | 246   |
| Core Block       | 323       | 142          | 465   |
| Admin Block      | 109       | 51           | 160   |
| Oncology Block   | 32        | 4            | 36    |
| Extra Mural      | 117       | 45           | 162   |
| TOTAL            | 1,197     | 350          | 1,547 |

These manpower figures are suitable for operational dashboards and workload context. They should not be used as individual app-user accounts.

## 9. Implementation and Verification Plan

| Phase                            | Work                                                                                                                   | Output / acceptance gate              |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| 1. Data import preparation       | Load the Excel sheets into staging tables; standardise aliases; retain original source references and Draft status     | No live app visibility                |
| 2. Block supervisor verification | Give each supervisor only their assigned block data to confirm floors, departments, rooms, landmarks and missing areas | Signed/recorded verification by block |
| 3. Routing verification          | Confirm primary/backup by shift, weekly off, leave replacement and escalation contacts                                 | Verified effective routing rules      |
| 4. Pilot configuration           | Enable selected verified blocks and departments; require landmark and photo; test push notifications and SLA           | Controlled NIMS Phase 1 pilot         |
| 5. QR expansion                  | Create stable QR location codes for wards, nursing stations, washrooms, labs and high-volume common areas              | Faster, lower-error ticket creation   |
| 6. Production rollout            | Activate remaining verified locations; publish SOP and train client/supervisor users                                   | Only Verified + active master data    |

### Minimum data required before production

- Official block naming and alias map.
- Verified floor and department mapping for pilot blocks.
- Supervisor employee code/email, app role, primary/backup assignment and weekly-off/leave coverage.
- FM category/subcategory ownership and priority/SLA policy.
- Authorised NIMS client users and department access.
- Resolution evidence, verification and reopen policy.

#### PRACTICAL PILOT APPROACH

Do not wait for a perfect architectural map of every room. Start with verified high-volume blocks and departments, require landmark + photo, monitor misrouted tickets, and progressively add exact locations and QR codes.

## 10. Appendix - Excel Workbook Guide

| Workbook sheet         | Contents                                                  | Primary use                                |
|------------------------|-----------------------------------------------------------|--------------------------------------------|
| Summary                | KPIs, source confidence, app-use flow and production gate | Management overview                        |
| Source Register        | Scope, limitations and status of each source              | Data governance                            |
| Location Master        | 327 draft indoor location records with draft IDs          | Primary app-import source                  |
| Morning Services       | 85 cleaned online service rows                            | Floor/room and schedule reference          |
| Evening Services       | 19 evening clinic/service rows                            | Cabin/floor reference                      |
| Deployment Departments | 238 block-wise department and post records                | Department dropdown and staffing analytics |
| Block Manpower         | Seven-block unskilled/semi-skilled summary and chart      | Management baseline                        |
| Staffing Adjustments   | 30 add/minus rows; total +42 / -22                        | Reconciliation reference                   |
| Supervisors & Roles    | 15 staff/role/shift/responsibility records                | User and routing preparation               |
| Routing Map Draft      | Proposed block/shift primary, backup and escalation       | Operational verification                   |
| Known Floors           | Floors explicitly mentioned and limitations               | Floor-master preparation                   |
| Data Gaps              | 16 open verification items with owner/priority            | Project tracker                            |
| App Import Mapping     | Source-to-Supabase table/field mapping                    | Development handoff                        |

### Source notes

ROLES.pdf pages 1-2 were used for names, shifts, responsibilities and assigned areas. Document 2.pdf page 1 supplied staffing adjustments; page 2 supplied block manpower totals; pages 3-11 supplied department deployment lists. The online block-wise services data was pasted by the user and cleaned for obvious OCR-like word insertions; it must still be checked against the original page and current NIMS operations.

#### FINAL ARCHITECTURE PRINCIPLE

The master data tells the system which supervisor should receive a ticket and gets that supervisor to the correct department. The exact landmark and photo tell the supervisor precisely where the problem is.